

Table 1: Static-promotion sensitivity (n=2, 16 threads, in-memory). Bold = workload’s argmax.

p_0	TPC-C (k TX/s)	Rnd-Read (k TX/s)	Skew-YCSB (M TX/s)
0 (off)	633 (−1.7%)	15 521 (−1.7%)	25.4 (−4.4%)
$\frac{1}{16}$	651 (+1.2%)	16 033 (+1.6%)	25.6 (−3.7%)
$\frac{1}{32}$	644 (paper)	15 786 (paper)	26.6 (paper)
$\frac{1}{128}$	622 (−3.3%)	16 016 (+1.5%)	26.0 (−2.0%)

Table 2: TPB vs the static $\frac{1}{32}$ default (n=5, 30 s, 16 threads). Welch two-sided $|t|$; 95 % significance needs $|t| > 2.31$. Throughput in thousands of TX/s; σ is sample standard deviation.

Workload	Variant	Mean	σ	Δ vs def	$ t $
TPC-C	default	634.4	13.3	—	—
	TPB	626.2	41.8	−1.29%	0.42
Rnd-Read	default	15 552	234	—	—
	TPB	15 815	558	+1.69%	0.97
Skew-YCSB	default	26 032	476	—	—
	TPB	26 561	373	+2.03%	1.95

Table 3: Promotion economics of the two B-tree page populations (uniform workload).

	inner nodes	leaf nodes
Count	~1 %	~99 %
Re-access rate	very high (every lookup)	near zero
Survives to reuse?	yes, millions of times	almost never (evicted first)
Promotion economics	benefit $\gg$ cost	cost only

Table 4: Role-aware promotion vs global constants on rndread (10 M-row single B-tree, 8 GB pool, n=5, 30 s, 16 threads). Bold = the significant win.

Config	k TX/s	Δ vs $\frac{1}{16}$	$ t $	Δ vs $\frac{1}{32}$	$ t $
global $\frac{1}{16}$ (best constant)	$15\,605 \pm 67$	—	—	—	—
global $\frac{1}{32}$ (paper default)	$15\,771 \pm 406$	+1.07%	0.90	—	—
inner $\frac{1}{16}$, leaf never	$16\,454 \pm 491$	+5.44%	3.83	+4.33%	2.40
inner $\frac{1}{16}$, leaf $\frac{1}{128}$	$15\,690$	+0.54%	1.01	—	—
inner $\frac{1}{8}$, leaf never	$15\,733$	+0.82%	0.84	—	—

Table 5: Effect of turning leaf promotion *off* (inner fixed at $\frac{1}{16}$) across workloads ordered by access skew. $\Delta = \text{leaf-off} - \text{leaf-on}$.

Workload (skew)	leaf-off	leaf-on	Δ	$ t $	Verdict
rndread (uniform)	16 454	15 690	+4.87%	3.28	leaf-off helps (sig.)
TPC-C (mild)	642	636	+0.91%	0.70	leaf-off helps (noise)
Skew-YCSB ($\zeta=0.9$)	25 842	25 959	−0.45%	0.25	leaf-off neutral

Table 6: Development arc of the adaptive controllers. Δ vs the static $\frac{1}{32}$ baseline.

Variant	Design	TPC-C Δ	Skew-YCSB Δ
CBP-v3	64-class hill climber, hot-path atomics	−11.5%	−22.4%
CBP-v4	thread-local counters + hit-rate bandit	−0.83%	−7.4%
TPB-v3	throughput signal, tol 5%, probe-every 8	−2.21% (sig.)	—
TPB-v5	throughput signal, tol 15%, probe-every 16	−1.29%	+2.03%